ELEC60022
ELEC70054

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART III/IV: MEng, BEng.and ACGI

MICROWAVE TECHNOLOGY

Wednesday, 10 December 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer question ONE. Answer TWO out of the remaining three questions.

Question ONE carries a weighting of 50%. The rest carry a weighting of 25% each.

This is an OPEN BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : K. Fobelets

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Special instructions for invigilators

This is an OPEN BOOK examination.

Special instructions for students

There are 4 (four) questions in total.

Question 1 is COMPULSORY and carries a weighting of 50%.

Answer 2 (two) out of the remaining 3(three) questions, each carrying a weighting of 25%.

All variables and abbreviations have their usual meaning.

1.

a) Parts of the frequency spectrum have limits on exploitation. For example:

i) As frequency decreases below 1 GHz, what affects the signal/noise ratio of a wireless communications system?

[4]

ii) Why is the frequency spectrum between 1 GHz and 10 GHz so convenient for commercial exploitation?

[4]

iii) What justifications are there for choosing 2.45 GHz as a suitable operating frequency for a microwave oven?

[4]

iv) Where in the frequency spectrum are the water and oxygen absorption peaks between 10 GHz and 200 GHz?

[4]

v) What is significant about the 38 GHz and 94 GHz frequency bands? Give an appropriate application for each band and state the reasons for choosing these applications.

[4]

vi) What is significant about the 60 GHz frequency band? Give two applications for this band and state the reasons for choosing these applications.

[4]

b) Lumped-element components exhibit undesirable frequency behaviour.

i) State the simple relationship between the unloaded Q-factor and frequency for a lumped-element tuned circuit. Define all variables clearly.

[4]

ii) Calculate the transit time cut-off frequency for a field-effect transistor having an intrinsic transconductance of 41 mS and gate-source capacitance of 0.2 pF. Does this cutoff frequency define the unity-gain bandwidth of an amplifier?

[5]

iii) What technique can be adopted to increase the Q-factor of a lumped-element tuned circuit using a simple transistor circuit? Give a basic rule-of-thumb, in terms of transit time cut-off frequency, for the frequency range that this technique can be applied to.

[4]

c) Distributed-element components exhibit undesirable frequency behaviour.

i) State the simple relationship between the electrical length and frequency for a transmission line operating in a (quasi-)TEM mode. Define all variables clearly.

[5]

ii) Which simple rule-of-thumb relates electrical lengths and physical lengths for integrated circuits that employ transmission lines? What are the practical implications of this as frequency decreases and increases?

[4]

iii) State the simple relationship between the attenuation coefficient's frequency response and frequency for an air-filled coaxial cable. Define all variables clearly.

[4]

2.

If the dielectric is Perspex, having a relative permittivity $\varepsilon_r = 2.6 - j\,0.018$ at 10 GHz, calculate the following parameters for a frequency of 10 GHz:

i) Q-factor. [3]

ii) Propagation constant. [4]

iii) Skin depth. [3]

iv) Thickness of a quarter-wave sheet. [3]

v) Power attenuation in dB per unit wavelength. [4]

vi) Intrinsic impedance. [4]

vii) Power flux density when the RMS electric field intensity is 1 V/m. [4]

3.

a) Describe the following terms for waves travelling on a lossless transmission line and in each case comment on their practical importance:

i) Velocity factor.

[3]

ii) Characteristic impedance.

[3]

iii) Return loss.

[3]

iv) Reflection coefficient.

[3]

v) Phase delay.

[3]

b) A lossless coaxial cable has inductance L [H/m] and capacitance C [F/m]:

i) Give expressions for the characteristic impedance and velocity factor for this line.

[2]

ii) Calculate the inductance and capacitance of a 1 m length of line if it has an impedance of 50 Ω and velocity factor of 0.6.

[2]

c)

i) Calculate the reflection coefficient and return loss for reflections at a load having an impedance of $75 + j\ 13$ Ω connected to a 50 Ω transmission line.

[1]

ii) Express the reflection coefficient in real and imaginary parts, and also as a modulus and phase angle.

[1]

iii) How does the reflection coefficient and the input measurement reference plane change as its distance to the load decreases?

[2]

iv) If the transmission line between the input port and the load decreases by quarter of a guided wavelength, what will the effect be on the reflection coefficient at the input measurement reference plane?

[2]

4.

An air-filled lossless metal-pipe rectangular waveguide has internals cross-sectional dimensions of $a \times b$ (where $a > b$). This waveguide operates in the fundamental TE_{10} transmission mode. where its propagation constant is γ_g. The propagation constants associated with its lowest-order cut-off frequency is γ_c and that of a TEM wave propagating in free-space is γ_0. These propagation constants are related by the following expression:

$$\gamma_g^2 = \gamma_0^2 - \gamma_c^2 \tag{4.1}$$

i) Using simple relationships, define γ_c in terms of the angular wavenumber, wavelength and broad wall dimension a.

[3]

ii) Using simple relationships, define γ_0 in terms of the angular wavenumber, wavelength and frequency.

[2]

iii) From (4.1) derive an expression for γ_g, below the cut-off frequency, in terms of broad wall dimension a and frequency. From this derivation, how would you describe this mode and how did you come to this conclusion?

[5]

iv) From (4.1) derive an expression for γ_g, above the cut-off frequency, in terms of broad wall dimension a and frequency. From this derivation, how would you describe this mode and how did you come to this conclusion?

[5]

v) If $a = 22.86$ mm, calculate the cut-off frequency f_c and the propagation constants at 1 GHz below f_c (i.e., at $f_0 = f_c - 1$ GHz)) and at 1 GHz above f_c (i.e., at $f_0 = f_c + 1$ GHz)). With both propagation constants, give the numerical values and their units.

[10]